

PAPER_699: Fornax Ultra-Deep Field: 10,000+ Galaxy Statistics and UQFF

Author: Daniel T. Murphy **Date:** 2025

Class: FornaxConstellationUHDF

CP4 Entry: #283

Keywords: Fornax HUDF, ultra-deep field, galaxy evolution, comoving volume, Schechter function, UQFF

Session: 174 | **Version:** v5.31

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Abstract

The Fornax constellation hosts one of the deepest Hubble observations, revealing 10,000+ galaxies across $z = 0.1$ to 6.5. Statistical tools including the Schechter luminosity function and UQFF galaxy evolution corrections are derived.

1. Survey Parameters

Parameter	Value
N_{gal}	$\sim 10,000$
Area	11 arcmin ²
z range	0.1–6.5
$\bar{z}$	1.8

2. UQFF Galaxy Number Evolution

$$N_{UQFF}(z) = N_{obs}(1+z)^{-1.5} \left(1 + \frac{\rho_{SCM}}{\rho_{UA}} \right) (1 + f_{TRZ})$$

3. Comoving Volume Element

$$\frac{dV}{dz} = \frac{c}{H(z)} D_c^2(z) d\Omega$$

4. Schechter Luminosity Function

$$\phi(L) = \frac{\phi^*}{L^*} \left(\frac{L}{L^*} \right)^\alpha e^{-L/L^*}$$

With $\alpha \approx -1.3$ for field galaxies.

5. Hubble Parameter

$$H(z) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$$

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$$\rho_{\text{vac}}(r) = \rho_{\text{vac, [SCm]}} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- Williams et al. (1996), AJ, 112, 1335 (HDF-S)
- UQFF Framework, Star-Magic Session 174

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